

Right (or) wrong → something universal or relative?

if something is wrong ⇒ wrong universally

Ex: Lying is wrong

↳ is that universally wrong?

⇒ sociology

studies specific cultures

What are Norms, Rules, Regulations to follow

customs, Traditions, Religions

* ~~Engl~~ philosophy → related to all other Ohhok!!

Science → starts with assumption ⇒ observation → knowledge
leads to

(Empirical)

Philosophy ⇒ questions ^{assumptions} ~~observations~~ itself.

Rationalists → pure reason (like Math) gives knowledge

Empiricists → true empirical observation → gives knowledge

What is philosophy??

→ philo ⇒ LOVE ; Sophia : WISDOM

philosophy → love for wisdom

Compare with science

Science → did many things

Technology, updated theory
Innovations etc

* relevance of
philosophy??

Why philosophy??

⇒ what philosophy does?

→ muddled in questions

↳ leads to other questions...

X

⇒ But it is foundation of any kind of Discipline

Doctorate: PHD ⇒ Doctorate & philosophy

Why philosophy is here?? NICE

When we do PHD, ⇒ We do foundational research

you go into foundations of our topic

→ Are actually we are doing philosophy
→ We don't need to study philosophy to do philosophy

→ Foundational Questions?

* Are space & time diff entities? (Einstein)

(Einstein) → His theory is not based on observations.
→ He is not doing empirical science

→ He is questioning the assumptions
(like what is time & what is space?)

He comes up with his turns
→ Quantum physics → open whole box of philosophical questions

Newton
Galileo
Kepler

Natural philosopher

Newton book ??

way to know the scientific truth

→ psychology → when response to question?
How to study the MIND ...

EX!

→ AI : philosophical project

WIMP

⇒ Why study philosophy?

→ The oldest surviving discipline!

→ learn to reason & argue & disagree

→ learn to not conform to readymade beliefs masquerading as truths

→ Dunning-Kruger effect:

Introduces Awareness of Dunning-Kruger effect

→ People with little knowledge often overestimate their understanding

→ The more ignorant we are, the more confident we might feel

Socratic irony

→ I don't know & proceed #

Socratic Irony (sd?)

* Socrates → practice questioning & admitting Ignorance to expose true wisdom.

** THE UNEXAMINED LIFE IS NOT WORTH LIVING

What is philosophical question?

Questioning of question

① Logic

⇒ How we think about thinking itself

(1) looks at that question

→ How we are to

⇒ crucial &

Normative (vs) Descriptive

* philosophy usually ⇒ deals with Normative Questions

facts (vs) values

⇒ Facts have to do with "What is the case" prescribing

⇒ Values have to do with "What are to be the case" ?

Describes

→ economics vs types

philosophy of economics → how you classified types of Economics

** Notion of power ? → making one school dominant

Connected to political theory

All includes both Normative & descriptive

any field value free ??

** What is Euclidean & Non-Euclidean maths?

Science ?
maths ?

* it is also involved in clarifying

concepts like

→ Justice, language, Meaning
Sport, truth knowledge etc

logical foundation

$$1+1=2$$

↳ 200 pages proof

→ philosophy questions - Assumptions.

↳ & the observation which lead to knowledge ~~is~~

Is math innovated? (or) ~~was~~ discovered

↳ In science & physics → many discovered mathematically

→ by making calculations ...

It.

* It involves providing Arguments to conclusion

→ A good argument gives reader a good reason to believe in conclusion.

→

History of philosophy =

Does observation really leads to knowledge??

↳ It is a history of powerful

arguments that stood the test of time

pick an activity ex: anything, sport, music, cooking, coding

& philosophize it

→ Deep questions

→ challenge / think about assumptions

→ Arguments &

→ reasoning etc

Build a thesis

ex:

Football ⇒

→ Game (or) sport

→ Just a game (or) any form of art, culture, Tradition.

Normative questions?

rich teams → buy good players?

How to play?

diver & fake injuries to win?

↳ Build a thesis

Question Assumptions?

Goal ⇒ win why?
it should be?

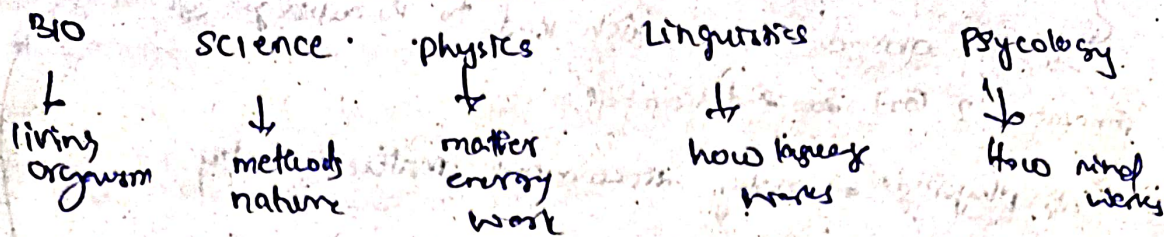
Philosophy (in its)

* → sticking to it but to UNANSWERED Questions.

⇒ philosophy as oldest discipline in Western thought

↳ means began as search for understanding everything
life, nature, ...

all fields. New specialized fields → broke from philosophy



Domains: ① Logic → right reasoning?
Study of valid arguments
Correct thinking

② epistemology
→ What is knowledge?
→ How knowledge possible
⇒ its Sources, Validity, Assumptions etc
→ Justified true belief plato

③ Ethics:
→ What is right (or) wrong?
Good or bad?
⇒ Morals, values, Justice, RIGHTS etc

④ Axiology
What is value?

⑤ Aesthetics
What is beauty?

Includes in all areas

Study of art,
taste,
music, beauty - 4
⇒ emotions ✕

* AI: → Getting these all philosophical questions back to
New philosophical questions

→ What is intelligence?
→ Can machines be conscious? etc
What you mean "think"?

Limitations?
Intelligent? → If should fit
rights they have?

If AI with intelligent → then come with some
Ethics #

all these what is thinking?
→ ✕ ✕ ohhoh!

→ it is reasoning the question?

(or) Giving most probable answer from Database. #

???

Argument → Ignorance & better of philosophy #

Thought experiments → crucial to philosophy

→ ex: what will happen if we go with speed of light?

Chinese → Chinese
** can you say the computer understands Chinese?

Trolley problem: → famous in ethics

→ self-driving cars

→ we need to make a decision (when it is out of control)

it is self driving car # → what it need to do?

Socrates → knew he didn't know

"All I know is I know Nothing" ** I know justice

I don't know anything → what is justice?

giving counter examples. → to people answers?

=> it is Socratic Method.

=> it is very efficient method of teaching # (way)

Examining your life → worthy #

disagree

Blindly following custom, traditions

authority => lesser of life

then a life which involves examining # philosopher

"unexamined life is not worthy"

→ questioning assumptions

→ providing arguments

→ Normative

→ Socratic method irony for effect

→ clarifying concepts

is it possible to reduce values $\rightarrow$ facts?
murder $\rightarrow$ look & tell 'it is made it wrong'

science

is it possible to travel backward in time?

protonize

$\rightarrow$ philosophical question

values:
Notion's apply to program
Time complexity
Readability
 $\rightarrow$ modularity

Bullshit

on Bullshit

Analysing concept of Bullshit

$\rightarrow$ what makes bullshit a Bullshit?

$\Rightarrow$ diff b/w Bullshit & Lying?

No care

Care for truth atleast

$\Rightarrow$ No care for
neither truth
nor lie

$\rightarrow$ atleast to make effort to
stick to that lie

$\rightarrow$ ~~consistency~~ cares for lie & truth

How do we

$\Rightarrow$ clarify notion of murder,
color,
Language?

Analysis of concepts

$\downarrow$
Also has Normative elements

$\Rightarrow$ clarifying in justice

$\hookrightarrow$ we are if justice means "to

this can't be termed as Justice

Free speech means this
that exists

Enables us to realize what
we don't know

Thesis: Better thesis

Synthesis:

Philosophy is Intrinsically Interdisciplinary

sci $\rightarrow$ phy $\rightarrow$ Bio

How Interd $\rightarrow$

how diff fields hanging $\rightarrow$ engineering hanging together with ethics

make artifacts

$\rightarrow$ embed certain values

$\rightarrow$ enabling/constraining
actions

because bad philosophy is all around ??

critical thinking
will see

L-2:

logic & critical thinking:

closest term comes to philosophy in INDIA: ?

→ Knowledge "Darshana" translates to philosophy

⇒ Darshana = way of view of world

⇒ Term which translates to critical thinking

→ Anvayashiki ??

Arguments, → Give reasons to in support of certain posn

⇒ Anyone reading Argument ⇒ should Ask it reasonable to believe in Truth of Conclusion #

⇒ Any research paper → Can be seen as (philosophically) Argument & Thought of as an Argument **

Economics → can observe

Neo is better than other

with Reasoning

What is logic?

→ called Arguments

⇒ The study of rational argumentation

to understanding the differences between Good Arguments & bad Arguments.

→ ~~what is~~ understanding what makes an Argument Good (or) bad?

Good argument — why it is Good ?

Ques

Bad " — why it is BAD ?

Youtube → influence

→ Es) gives Arguments.

Fallacies ⇒ Logical Illusion

→ it seems like a good argument but it's not really a good argument #

① comprehensiveness it should be comprehensive of pros & cons

② clarity (clear) → No ambiguity

③ There should be justification

⇒ chain of logical reasoning to every area

→ No floating

we fall for emotions
Rationalists

if purely rational beings ⇒ very little scope to have false beliefs

Good

⇒ A Normative discipline

→ it will tell us "How we should reason"
rather than How we do reason

~~Normative~~
Normative
(vs)
Descriptive

→ The art of self-defense what makes proper reasoning

⇒ To immune ourselves from Arguments which are fallacies

⇒ Against bad arguments

we don't fall for lower hanging fruits in terms of Beliefs, products people

⇒ Captain of own ship (master/director of movie)

⇒ By other reasons to give value for logic: ??

Normative: Concerned with "How things ought to be"

Descriptive: Concerned with "How things are",

focus on ideal standards

simply observes real world arguments without judging them.

What makes an argument valid

as self-defence

→ Against bad arguments

Emotional persuasion & electronic

propaganda & non-reasonable methods

⇒ Helps to

→ Support conclusions with strong, well-grounded premises

* Build Arguments that withstand criticism

* write clearly

(Gives reasoning to all critiques)

Clarity & Effective Writing #

~~Ex: The reader~~ Thesis → is an argument

→ it supporting a certain conclusion

⇒ ∴ Why the reader should believe your conclusion?

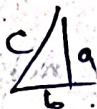
What reasons have to believe **

What makes a Argument Good ⇒ Inference ??

In a Good Argument, The reader should be able to Infer the reasoning

(like math proof)

ex: $a^2 + b^2 = c^2$



⇒ 2 parallel lines can't meet never

⇒ [deduce something from AXIOMS]

∴ reader also can inference in same way **

Inferences in all time: (in everyday life)

* See them in earphones

→ Infer they may be listening to music

what grounds that inference

⇒ Experience

↳ from Deductive

∴ Two types Inductive & Deductive

ex

science

Experience

→ Maths

→ only earphones we use
∴ someone also → listen music

we put on earphones → not to disturb

others → ∴ not to disturb (can).

∴ There is probability involved in Induction.

↳ it is not always certain.

→ sincerely ans. → gives probably truth

→ it is different from Deductive reasoning

Deductive reasoning

↳ This is always certain

→ correct answer

IIT studying
engineering

→ JEE ✓ Deductive

Grounds the truth

good Hindi → note

two people → couple

bioinform → happy

} Inductive

↳ what makes these inferences

Good?

⇒ subjectively

→ Fallacy of Generalisation

⇒ Delhi

rough & tough

X (may be wrong)

⇒ Ex: of Deductive reasoning:

I have
10 rupees
in pocket

5 rupees chai
bought

∴ remains 5

→ deductive #

⇒ Anything involving "Transactions"

→ deductive #

Formal & Informal logic

Formal:

↳ it has to do with "Structure" of arguments

(or)
"form"

→ Form of argument → evaluated for ~~formal~~ Good or bad

Deduction:

it is form of an argument, which makes an argument valid

→ Not content

⇒ All men are mortal

⇒ All chairs are mortal X

Logic
↕
Practice

Formal logic $\rightarrow$ Has to do with "FORM" & "STRUCTURE" of arguments

$x \xrightarrow{S} P$ $\therefore$ $x \rightarrow P$ $\quad \quad \quad \text{All } S \rightarrow P$

one form of argument

$\Rightarrow$ syllogism

premise 1 :

premise 2 :

Conclusion:

$\rightarrow$ Anything for S, x, P

$\Rightarrow$ always valid argument

$\Rightarrow$ valid regardless of content

* whatever x, S, P always true *

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All $S \rightarrow P$

All $Q \rightarrow P$

therefore $Q \Rightarrow S$ *

& it valid ? X

All dogs ~~are~~ mammals

All humans mammals

P : mammals

Q : human

S : dogs

$Q \Rightarrow S \Rightarrow$ All dogs are humans X

Consists of ① truth tables

② Predicate logic

③ categorical propositions

④ syllogisms

square of ⑤ propositional logic

develop various ways of logic

$\Rightarrow$ categorical logic

All $S \rightarrow P$

no $S \rightarrow P$

some $S \rightarrow P$

& some $S \neq P$

four kinds of statements

diff relationships

square of a position



$\Rightarrow$ GATES

[logic gates]

✓ truth tables

Informal logic:

$\rightarrow$ Has to do with "Content or Meaning" of premises in Argument & proposition

Gameplay everyday arguments

* can't be neatly set into categories.

$\rightarrow$ There is lot of Meaning involved

$\Rightarrow$ Ex: Nobody is perfect $\Rightarrow$ I am Nobody
then I am perfect

it seems like an valid Argument *

$\rightarrow$ playing on word Nobody

$\rightarrow$ changing meaning of 'Nobody' b/w premises

- ① premise 1: I am Not perfect
 ② premise 2: I am Not perfect
 ③ conclusion: I am perfect
- nominal → noun
 {shift ↓ in meaning} X Follows

→ More to do with Identifying the fallacies in everyday reasoning.

∴ Need to do with content or meaning

→ lot of work going with LLM, Identify informal fallacies ^{Can they??}
 because they are not structured

they are based on meaning #

can they understand #

* inquiry Human task to try to understand the ~~polymorphic~~ nature of language:

~~polymorphic~~
Multiple meanings

Ex: BANK

→ 'n' mean

→ many bank

→ bank of river

→ banking of aeroplane

?? NICE

∴ forms a Major part of
"CRITICAL THINKING"

Articulating Disagreement:

Copplestone & Russell

① → does existence of God need to be understood by term "God"?

→ I presume → Supreme personality & distinct from world & creature of or

② Yes

③① → his posn is affirmative there is entity that exists

② no I should be "my posn" is Agnostic

I can't prove non-existence

but can't believe EXISTENCE

What is an Argument?

A good Argument

→ one which premises gives a "Good" reason to believe in conclusion #

Good reason ?? → It means Giving us Good Inferences

→ enable us to make give solid / strong inferences

→ It is INDUCTIVE ~~may be (might be) banker~~ X

If all bankers are ~~have~~ rich

If Tom banker → rich ✓ Tom is a banker

Invalid deductive Argument (diff) Inductive argument

(V)

Here it is Invalid Deductive Argument

Truth preserving!

Deductive Argument preserves truth

All DP's are CS students

All DP's take IHS

All IHS are CS Students

→ it is not possible for premises to be true & conclusion is false #

A valid Deductive argument exist ?? X Max NO #

Not logically possible

In a valid Deductive Argument ⇒

If premises are True ⇒ Conclusion must true #

(has to be)

In Deductive:

If conclusion observed false

at least

we can tell at least one of premise is false #

**

All math proofs are deductive

① Agree with AXIOMS

② inferred them correctly

③ prove is valid

→ what you proven is conclusive #

Science → Go back & Question them

→ some other phenomenon can change

Inductive: Tiny new phenomenon can falsify Einstein's theory

Deductive: → ~~a new~~ No amount of new evidence can falsify true conclusion

Inductive arguments are always $\rightarrow$ based on new evidence

$\therefore$ more significant $\Rightarrow$ (deductive)

Deductive argument Evaluate:

can be conclusion false if premises are true $\#$

~~if~~ $\#$ conclusion false

$\Rightarrow$ one of at least premise false

$\Rightarrow$ Invalid Argument

check



even in legitimacy

new discoveries $\Rightarrow$ lead certain premises lead to be false $\#$

$\#$ if it is possible that premises of deductive argument could be conclusion

from Inductive Argument.

all heavenly revolve around earth

sun is heavenly body

sun around earth

$\#$ may be conclusion of Inductive argument

not concerned by it

The basic form $\#$

No reality $\#$ in premises
only taking one $\#$

$\Rightarrow$ Types of Inferences:

Deductive:

$\rightarrow$ conclusion supported by its premises CONCLUSIVELY.

if premises are true $\Rightarrow$ conclusion has to be true $\#$

Good Deductive Argument is TRUTH-PRESERVING (When valid)

$\rightarrow$ an argument is truth preserving if

it is not logically possible for its premises to be true, $\#$
while its conclusion is false, $\#$

Logical vs physical Impossibility ?!

11/29 $\rightarrow$ valid Deductive Argument
but an unsound argument

} because one of the premises is false $\#$

come back ?

logical possibility:

Example
Tom → ① Deductive 12/29

if Tom is a banker } premises
if All bankers rich }
then Tom is rich } conclusion.

conclusion
Identify
↓
wrong
different

Conclusive

Therefore }
then } Conclusion
Hence }

② to is it logically possible?

is it guarantee

Likelihood

Inductive

if
All smokers die with cancer
Sue is smoker
Sue will die with cancer

Deductive

OK OK!!

③

④ Deductive

⑤ ~~Statistics~~

Statistics
empirical

Samples

Deductive
but invalid

⑥ Inductive

all are may not examined

Sample size

⑦ Deductive

If A then B

if not B then not A

Flowchart:

Deductive Argument

if premises true then it is impossible to be conclusion false

Yes

No

↓
valid

↓
Invalid
Unsound

All premises true

Yes

NO

SOUND

UNSOUND

Induction: → Tomorrow will be like today

→ when new evidence → change premises truth-value

INDUCTIVE

logical possibility (vs) physical possibility



↳ if something INCONSISTENT with the Laws of Nature

Something
INCONSISTENT with
laws of Logic

SCIENCE → tell us which is
physically impossible

Married Bachelor X

square circle X

RED & Not-red X

OXYMORONS are logically impossible

→ less is more

→ same difference

Laws of Logic/Thought:

① Law of non-contradiction

⇒ Cannot be both A & not-A

Ex: Nothing can both have property to & lack at same time

⇒ can't be both true & false

② Law of Identity:

(A = A)

if it is true, it is true

if it is false, it is false

⇒ Everything is identical to itself

③ Law of excluded middle: (either A (or) Not-A)

contradiction

in a valid deductive arg

⇒ premises true

conclusion false

⇒ leads to invalid

∴ contradiction

logically possible but physically impossible?

→ Archimedes

→ Give me a lever long enough & fulcrum on which to
place it & I shall move world!

20/29

① → valid deductive unsound

② → Inductive

once / thr. ⇒ like data empirical

Weak (or strong) $\Rightarrow$ Weak

Strong inductive:

in which ~~premise~~ premises makes conclusion more likely $\#$
 $\Rightarrow$ premises truth of premises $\rightarrow$ makes $\rightarrow$ truth of conclusion
very probable $\#$ (highly probable)

Correlation
Causality

③ ~~Reductive~~
INDUCTIVE (Weak)

Weak ??

Eat other Junkfood $\rightarrow$ & drink tea
No loss

$\therefore$ Inductive

$\hookrightarrow$ green tea contributes to weight loss

Contribution is $>>$ causation

④ Strong
INDUCTIVE

Chances

21/29 :

① Reductive

Valid Sound

Invalid